Deliverable 4: Assessment Report

AI-powered procurement intelligence for medical laboratory and diagnostic equipment in World Bank-financed health projects: what was built, what the data shows, and what comes next

Item	Detail
Run date	22 October 2026
Domain	Medical laboratory and diagnostic equipment in World Bank-financed health projects
Data	public World Bank APIs only: Major Contract Awards, Procurement Notices, Projects, Documents & Reports, WDI (CPIA, WGI)
Coverage	2,220 lab and diagnostic contracts (US$1,375.4M), 91 countries, 179 projects, FY2015 onwards
Companion deliverables	1 Data Model · 2 Approach (technical proposal) · 3 Prototype (pipeline + chatbot)

1. Summary

World Bank health projects buy a large amount of laboratory and diagnostic equipment, but the records of what was bought, how and why it worked sit in separate places: contract award data, procurement notices, and PDF reports. We built a working prototype that connects them for this one domain, across every country. It turns them into answers a procurement team can use, through a cited chatbot and package briefs.

Question	Answer from the prototype
How big is the domain in public data?	**2,220 contracts worth US$1,375.4M** across 91 countries, 179 projects and 1,277 suppliers, found by classifying 17,058 health-goods contract descriptions in four languages
Can the silos be joined?	Yes. **100%** of contracts link to their project and **67%** to their tender; 131 project reports connect through Project ID (115,327 graph links)
What does the market look like?	Diagnostic imaging is the largest category (US$451.6M). 53% of value was signed in 2020-2022 (COVID-19). Domestic suppliers took 58% of value; 16 suppliers won in two or more countries
How long do tenders take?	Medians from invitation to award range from 90 days (Laboratory analyzers) to 183 days (Biosafety, sterilization and cold chain), censoring-aware
Is the forecast better than a benchmark?	Yes. Out-of-time median error **47 days** vs 56 for cohort benchmarks (**17% better**)

Question	Answer from the prototype
Does the chatbot answer correctly?	It routed **100%** of 16 labelled questions to the right tool; ranked and aggregate answers matched an independent query in **6 of 6** checks; 100% of statements cited
Can hallucination be controlled?	Yes, by design. 100% of 360 brief statements cite evidence; the verifier blocked **124 of 125** deliberately injected false statements
Are the risk models ready to drive decisions?	Not yet. Single-bid AUC 0.67; re-tender skipped: not enough positives; award amendment AUC 0.55. Use as indicators only

2. What was built

Module (assessment §4)	Prototype component	Status
1. Ingestion and knowledge representation	contract-first domain selection, multilingual equipment taxonomy, entity resolution, Gold star schema, knowledge-graph edge table, section-aware PDF chunking	built and run
2. Predictive and prescriptive engines	Kaplan-Meier benchmarks by equipment type, gradient-boosted P50/P80 model, single-bid / re-tender / amendment classifiers, hybrid retrieval, package briefs	built and run
2. Retrieval for users	**chatbot** (`python -m streamlit run procurement_demo/app.py`): SQL for prices, suppliers and spend; benchmarks for timelines; retrieval for lessons; graph diagrams for relationships	built, run and evaluated
3. Guardrails and attribution	evidence registry, verifier (citations, numbers, quotes, pages), rating-conflict resolver incl. IEG reviews	built, run and red-teamed
4. Continuous learning and rollout	incremental ingestion, monitoring and retrain gate, feedback tables, KPI view	designed and written as Databricks notebooks 21-29; **not yet executed** (trial workspace blocked job runs)

3. Data: how the domain was assembled

Step	Result
Health-practice Goods contracts pulled (FY2015+)	17,058
Classified as laboratory or diagnostic equipment	2,220 contracts, US$1,375.4M

Step	Result
Projects that bought such equipment	179 (all their 14,613 health-goods contracts kept as market context)
Procurement notices of those projects (Goods)	15,158 → 10,403 tenders (1,947 for lab and diagnostic items)
Project documents parsed (PAD, ISR, ICR, IEG review, procurement plan)	131 → 6,187 section-aware passages
Suppliers	7,835 raw names → 7,540 resolved (94 fuzzy merges)

Data quality found along the way.

- **Language and encoding.** Descriptions come in English, French, Spanish and Portuguese, and accents often arrive garbled ("r?actifs"). Patterns therefore tolerate a wildcard in place of accents.
- **Uninformative descriptions.** Some contracts are described only as "Lot 3", "Item 42" or "no contract". They stay unclassified rather than being guessed.
- **Missing invitation notices.** Many emergency purchases (direct contracting, UN agencies) have no invitation notice. They count in market statistics but cannot give a tender duration. Only 1,068 tenders have both an invitation and an award; 122 are still open (censored) and 442 had no award after two years.
- **No completion reports yet.** Most COVID-era projects are still active, so lessons come mainly from older, closed health projects (150 lessons extracted).

4. What the data shows about laboratory and diagnostic procurement

4.1 Spend by equipment type

Equipment type	Contracts	Value	Share	Median contract	Countries	Suppliers	Top-5 supplier share	Domestic supplier share
Diagnostic imaging	292	US$451.6M	33%	US$565,485	61	216	27%	62%
General laboratory and diagnostic equipment	427	US$264.1M	19%	US$150,000	66	342	44%	44%
Molecular diagnostics (PCR, GeneXpert)	498	US$212.2M	15%	US$77,380	60	227	23%	70%

Equipment type	Contracts	Value	Share	Median contract	Countries	Suppliers	Top-5 supplier share	Domestic supplier share
Reagents and cartridges	428	US$151.2M	11%	US$84,248	52	291	31%	45%
Biosafety, sterilization and cold chain	226	US$126.2M	9%	US$68,136	55	189	53%	57%
Rapid tests and test kits	172	US$117.1M	9%	US$340,175	46	107	57%	62%
Laboratory analyzers	119	US$35.4M	3%	US$132,000	32	99	40%	81%
Microscopy	34	US$10.0M	1%	US$116,964	17	28	72%	97%
Mobile and modular laboratories	15	US$6.9M	1%	US$174,374	5	13	83%	68%
Laboratory information systems	9	US$0.7M	0%	US$39,124	3	8	95%	54%

Top-5 supplier share is the share of the type's value won by its five largest suppliers; a high value points to a concentrated market or closed systems. *Domestic* means the supplier is registered in the borrower country.

4.2 Where and when

Region	Contracts	Value	Share	Countries
Sub-Saharan Africa	868	US$503.0M	37%	35
Europe & Central Asia	289	US$319.6M	23%	17
South Asia	566	US$210.9M	15%	5
Middle East, North Africa, Afghanistan & Pakistan	138	US$119.0M	9%	7
East Asia & Pacific	115	US$103.2M	8%	12
Latin America & Caribbean	178	US$101.3M	7%	15

Top countries	Value	Share
India	US$166.3M	12%
Turkiye	US$75.6M	5%
Madagascar	US$66.4M	5%
Serbia	US$64.6M	5%
Uzbekistan	US$56.2M	4%
Ukraine	US$44.5M	3%
Ghana	US$42.5M	3%
Iraq	US$42.0M	3%
El Salvador	US$34.9M	3%
Senegal	US$31.2M	2%

Signing year	Contracts	Value
2019	38	US$39.4M
2020	805	US$381.7M
2021	310	US$177.1M
2022	212	US$166.0M
2023	326	US$262.0M
2024	251	US$192.1M
2025	184	US$111.9M
2026	67	US$45.1M

Spending peaked in 2020; 53% of all value was signed in 2020-2022.

4.3 Who supplies

Supplier	Based in	Contracts	Value	Share	Countries	Main equipment types
UNITED NATIONS OFFICE PROJECT SERVICES (UNOPS	World	23	US$146.0M	11%	9	General laboratory and diagnostic equipment, Molecular diagnostics (PCR, GeneXpert)

Supplier	Based in	Contracts	Value	Share	Countries	Main equipment types
MAGNA PHARMACIA	Serbia	3	US$37.3M	3%	1	Diagnostic imaging
UNITED NATIONS DEVELOPMENT PROGRAMME (UNDP)	World	8	US$36.0M	3%	3	General laboratory and diagnostic equipment, Reagents and cartridges
BIOSENSOR HEALTHCARE	India	38	US$32.1M	2%	1	Rapid tests and test kits
UNITED NATIONS CHILDREN'S FUND (UNICEF)	World	12	US$27.9M	2%	9	Rapid tests and test kits, Reagents and cartridges
WORLD HEALTH ORGANIZATION (WHO)	World	6	US$25.7M	2%	4	General laboratory and diagnostic equipment, Molecular diagnostics (PCR, GeneXpert)
Philips Medical Systems Nederland B.V.	Netherlands	5	US$23.0M	2%	2	Diagnostic imaging
GE MEDICAL SYSTEMS	France	8	US$23.0M	2%	3	Diagnostic imaging
MEDICOM D.O.O. ŠABAC	Serbia	3	US$17.9M	1%	1	Diagnostic imaging
VIMA (VISION MADA)	Madagascar	1	US$12.7M	1%	1	Biosafety, sterilization and cold chain
APELEM	France	3	US$12.5M	1%	2	Diagnostic imaging
SIEMENS HEALTHCARE PVT. LTD.	India	11	US$11.8M	1%	1	Molecular diagnostics (PCR, GeneXpert), Reagents and cartridges

16 suppliers won lab or diagnostic contracts in two or more countries. They are the firms where cross-country evidence (price, delivery, after-sales) is most useful.

4.4 How it was bought

Procurement method	Contracts	Value	Share
Direct Selection	737	US$574.4M	42%
Request for Bids	730	US$540.6M	39%
Request for Quotations	733	US$218.4M	16%
UN Agencies (Direct)	20	US$41.9M	3%

Year	Direct / single-source share of value	UN agency share of value
2019	24%	0%
2020	64%	11%
2021	57%	18%
2022	26%	18%
2023	44%	29%
2024	33%	17%
2025	28%	21%
2026	26%	0%

4.5 How long tenders take

Kaplan-Meier estimates of days from invitation to award, by equipment type (all countries; still-open tenders counted as censored):

Equipment type	Tenders	Awarded	Median days	P80 days	Median if open tenders ignored
Laboratory analyzers	13	13	90	141	90
Molecular diagnostics (PCR, GeneXpert)	32	29	122	222	114
Rapid tests and test kits	14	13	123	180	110
Diagnostic imaging	70	64	134	247	124
General laboratory and diagnostic equipment	84	70	145	288	112
Reagents and cartridges	30	27	150	253	149

Equipment type	Tenders	Awarded	Median days	P80 days	Median if open tenders ignored
Biosafety, sterilization and cold chain	30	27	183	261	179

5. Results by module

5.1 Module 1: one connected knowledge base

Measure	Result
Contract → project link rate	100%
Contract → tender link rate	67% (9,600 exact + 182 fuzzy)
Supplier resolution	7,835 names → 7,540 suppliers
Lessons extracted	150 (68 procurement-related), 98.0% quote-verified
Knowledge-graph edges	115,327

5.2 Module 2: models

Out-of-time validation: trained on tenders opened before 2024-01-01, tested on later ones. All health-goods tenders of the domain projects are used for training, with equipment type as a feature.

Model	Result	Baseline	Train / test
Days to award, P50	MAE **47 days**	KM benchmark 56; single median 73	759 / 259
Days to award, P80	coverage **0.76**	target 0.80	
Single responsive bid	AUC **0.67**, top-decile lift 2.6×	base rate 0.417 → 0.281	772 / 295
Re-tender	skipped: not enough positives	base rate 0.005	1,163 / 469
Award amendment (proxy)	AUC **0.55**, top-decile lift 1.0×	base rate 0.391 → 0.447	773 / 295

Main drivers of tender duration: `bid_window_days`, `method_code`, `review_type`, `country_iso3`, `env_category`, `major_sector`, `wgi_control_of_corruption`.

5.3 Module 2: retrieval

Risk topic	Relevant lessons	Precision@5	Precision@10	Recall@20	Random
delay	61	0.80	0.70	0.65	0.41
capacity	80	1.00	0.90	0.85	0.54
fiduciary	12	1.00	0.70	0.83	0.08
specification	4	0.20	0.20	0.50	0.03

Structured-to-unstructured check: a project's own tender descriptions retrieve its own reports in the top 10 for **88%** of 16 projects (MRR 0.58).

5.4 Module 3: guardrails and conflicting evidence

Over 25 briefs for real held-out packages, 360 statements were produced, **100%** with valid citations. Five kinds of false statement were injected into every brief; the verifier blocked **124 of 125 (99.2%)**.

Injected statement	Why it was blocked
Among the 53 most similar past Goods tenders, the most common method was Request for Bids (7 of 12); median nu…	unsupported numbers [53.0]
“Contractors consistently delivered ahead of schedule across all lots.”	quote not in source: Contractors consistently delivered ahead of schedule across
Framework agreements always reduce procurement time by half.	no citation
Among the 12 most similar past Goods tenders, the most common method was Request for Bids (7 of 12); median nu…	unknown citation ['L999']; unsupported numbers [12.0]
“Introduction of c ontract terms against non- deliveries and poor deliveries .” (ICR, P152625, p. 80) - note: …	page 80 not in cited evidence

Conflicting evidence. 34 projects have dated ratings from several sources (ISR supervision, the Bank's ICR, IEG's independent review). In **8** of them the sources fall on opposite sides of the satisfactory line. Example: P129539 reached *Moderately Unsatisfactory* during supervision; the final rating, from the ICRR, is *Satisfactory*. The resolver keeps the most authoritative source as the headline (IEG PPAR > IEG ICR Review > ICR > latest ISR), stores the full trajectory, and the chatbot and briefs cite both sides.

6. The chatbot

Question	Expected tool	Chosen	Route ok	Exact	Statements	Dropped by verifier
Which supplier won the cheapest GeneXpert or PCR contract?	rank_low	rank_low	yes	yes	6	0
What is the largest diagnostic imaging contract?	rank_high	rank_high	yes	yes	6	0
What are the lowest value laboratory reagent contracts in Nigeria?	rank_low	rank_low	yes	yes	3	0
Who are the top suppliers of laboratory analyzers?	suppliers	suppliers	yes	yes	7	0
Which vendors supplied microscopes since 2019?	suppliers	suppliers	yes	-	7	0
How much was spent on lab and diagnostic equipment by country?	aggregate	aggregate	yes	yes	7	0
What is the total value of rapid test kit contracts?	aggregate	aggregate	yes	yes	2	0
How long does it take to award a PCR tender?	timeline	timeline	yes	-	2	0
How long do laboratory analyzer tenders take from invitation to award?	timeline	timeline	yes	-	2	0
Why were laboratory equipment deliveries delayed?	lessons	lessons	yes	-	8	0
What lessons did projects learn about maintaining diagnostic equipment?	lessons	lessons	yes	-	8	0
What risks should I plan for when procuring X-ray machines?	lessons	lessons	yes	-	9	0
Show the entity relationship diagram	graph	graph	yes	-	9	0
How is Kenya connected to its lab equipment suppliers?	graph	graph	yes	-	4	0

Question	Expected tool	Chosen	Route ok	Exact	Statements	Dropped by verifier
Tell me about project P173836	project	project	yes	-	10	0
What did UNITED NATIONS OFFICE PROJECT SERVICES (UNOPS) supply?	supplier	supplier	yes	-	5	0

Route accuracy **100%**, exact answers **6/6**, 0.06 s per question on a laptop. Sample answers (verbatim from the run):

> **Q: Which supplier won the cheapest GeneXpert or PCR contract?**
> *tool: rank_low · 6 evidence items*
>
> - The lowest-value contract for Molecular diagnostics (PCR, GeneXpert) went to SCI-FI BIOLOGICALS (India): contract 1696951, US$34, for “Primers and Probes [5 nos]”. `C1`
> - AVON PHARMO CHEM (PRIVATE) LIMITED: US$126 (contract 1697389, Sri Lanka). `C2`
> - M.M. SUPPLIERS: US$131 (contract 1696989, India). `C3`
> - LABSERVICE: US$176 (contract 1665197, Kyrgyz Republic). `C4`
>
> **Q: Who are the top suppliers of diagnostic imaging equipment in Sub-Saharan Africa?**
> *tool: suppliers · 2 evidence items*
>
> - 71 suppliers won 84 contracts worth US$104,305,138 for Diagnostic imaging in Sub-Saharan Africa. `D1`
> - AFRIMED MEDICAL SUPPLIES LIMITED ACCRA GHANA (Ghana): 1 contract, US$9,382,558 (9% of value), 1 country. `D1`
> - ELSMED (Cote d'Ivoire): 1 contract, US$8,169,922 (8% of value), 1 country. `D1`
> - EXP-MEDICAL SYSTEMS AFRICA LIMITED (Ghana): 1 contract, US$6,999,643 (7% of value), 1 country. `D1`
>
> **Q: How long does it take to award a laboratory analyzer tender?**
> *tool: timeline · 1 evidence items*
>
> - Laboratory analyzers: half of tenders are awarded within 90 days; 80% within 141 days (n=13). `B1`
> - Medians are censoring-aware: tenders still open count as at least their current age, so slow tenders are not silently dropped. Plan milestones on the P80, not the median. `B1`
>
> **Q: What problems did projects have with laboratory equipment maintenance?**
> *tool: lessons · 12 evidence items*
>
> - “These conditions were the adoption of the: (a) Cancer Control Improvement Program with Action Plan by the Government in July 2020; (b) rulebook for the establishment of a Medical Equipment Maintenance Unit within the MoH by the Government in October 2019 , and (c) issuance of a Decree to adopt the H” (ICR, P129539, p. 30) - note: this project's supervision ratings fell to Moderately Unsatisfactory before the final ICRR rating of Satisfactory, so the lesson reflects a recovery. `L1` `R1`
> - “In addition to the change in the MoH leadership in mid -June 2020, the COVID -19 pandemic also slowed down and adversely affected implementation progress and delays were also encountered for the

> project restructuring, for the initiation of the agreed procurement processes and subsequent partial canc" (ICR, P152625, p. 31) - note: this project's ratings were mixed across sources, and the final ICRR rating is Moderately Unsatisfactory, so the lesson comes from a project that underperformed. L2 R2
> - "At the Government's request, the WB activated the CERC and reallocated USS$14 million to expedite implementation of the Master Plan." (ICR, P157291, p. 27) L3
> - "Notwithstanding the cancellations, the project was still able to achieve the objectives." (ICR, P131825, p. 36) - note: this project's supervision ratings fell to Unsatisfactory before the final ICRR rating of Moderately Satisfactory, so the lesson reflects a recovery. L4 R3

7. Example output: a brief for a new package

Input: *Supply, installation and commissioning of automated real-time PCR systems and GeneXpert analyzers for regional public health laboratories, Lot 2* (Goods, RFB, KEN, est. US$2,500,000).

Schedule

- Historical benchmark (Molecular diagnostics (PCR, GeneXpert) tenders using RFB (all countries)): half of tenders reach award within 124 days of the invitation and 80% within 240 days. [B1]
- Counting only completed tenders would understate the median by 6 days, because slow tenders are still open. [B1] [D1]
- For this package the model estimates 155 days to award (P50) and 252 days at the 80th percentile. [M1]
- Plan the award milestone for 08 Nov 2027 or later (planned invitation 01 Mar 2027 plus the P80 estimate) rather than the median. [D2] [M1]

Risks

- Expected cycle is longer than the benchmark median (155 vs 124 days). [M1] [B1]

Market approach

- Among the 12 most similar past Goods tenders, the most common method was Request for Bids (7 of 12); median number of bidders 1.5. [T1] [T2] [T3] [T4] [T5] [T6] [D3]

Lessons from completed projects

- "In addition to the change in the MoH leadership in mid -June 2020, the COVID -19 pandemic also slowed down and adversely affected implementation progress and delays were also encountered for the project restructuring, for the initiation of the agreed procurement processes and subsequent partial canc" (ICR, P152625, p. 31) - note: this project's ratings were mixed across sources, and the final ICRR rating is Moderately Unsatisfactory, so the lesson comes from a project that underperformed. [L1] [R1]
- "Introduction of c ontract terms against non- deliveries and poor deliveries ." (ICR, P152625, p. 39) - note: this project's ratings were mixed across sources, and the final ICRR rating is Moderately Unsatisfactory, so the lesson comes from a project that underperformed. [L2] [R2]

- “The COVID -19 pandemic disrupted and delayed project activities.” (ICR, P149340, p. 19) - note: this project's supervision ratings fell to Moderately Unsatisfactory before the final ICR rating of Satisfactory, so the lesson reflects a recovery. [L3] [R3]
- “The COVID-19 pandemic significantly diverted resources and attention, delaying some project activities and necessitating adjustments to Project priorities.” (ICR, P152799, p. 31) - note: this project's supervision ratings fell to Moderately Unsatisfactory before the final ICRR rating of Moderately Satisfactory, so the lesson reflects a recovery. [L4] [R4]
- “At the Government’s request, the WB activated the CERC and reallocated USS$14 million to expedite implementation of the Master Plan.” (ICR, P157291, p. 27) [L5]
- “Future similar health projects should consider the lessons learned from this project regarding timely disbursement of county allocations.” (ICR, P152394, p. 37) [L6]

Supporting evidence

- “For each contract to be financed by the loan, for the different procurement methods or consultant selection methods, the need for prequalification, estimated costs, prior review requirements, and time frame will be approved by the Bank with the No Objection to the Procurement Plan.” (PAD, Procurement, p. 68) [P1]
- “For effective real-time monitoring and follow up of contract management plans for key high -value contracts under the project, the Bank team advised the PMU/MoH to establish and use a basic contract management tool.” (ICR, B. Environmental, Social, And Fiduciary Compliance, p. 34) [P2]
- “These delays did not leave time for the related procurement and, finally resulted in a decision to cancel US$34.4 million from the loan.” (ICR, B. Key Factors During Implementation, p. 32) [P3]
- “Performance on procurement arrangements and compliance was hindered by difficulties and delays encountered by the countries and regional institutions.” (ICR, B. Environmental, Social, And Fiduciary Compliance, p. 36) [P4]

8. Limitations

Limitation	Effect	What would fix it
Public data only	no bid prices per unit, technical evaluation reports, service contracts or equipment downtime	Phase 2: STEP procurement records, asset registers, maintenance logs
Keyword taxonomy	a residue of contracts stays "Other goods"; a few generic items are misclassified	labelled set of about 500 descriptions; LLM classifier for the residue
Few completion reports for COVID-era projects	lessons skew to older projects	add IEG thematic evaluations and the next ICR cohort as they publish
Rule-based lesson extraction locally	lessons limited to numbered paragraphs in Lessons / Key-factor sections	LLM extraction with quote check (notebook 24)

Limitation	Effect	What would fix it
Thin risk-model labels	single-bid, re-tender and amendment models are indicative only	more countries and internal records; recalibration
Verifier misses a fabricated number that appears elsewhere in the evidence	1 of 125 injected errors can pass	number-unit binding + entailment judge
Databricks path not executed	continuous-learning loop designed, not measured	run notebooks 21-29 when the workspace allows jobs

9. Recommendations and next steps

1. **Pilot the chatbot with procurement specialists and biomedical engineers** in two or three health programmes that are about to buy lab equipment. Measure plan-preparation time and evidence usefulness.
2. **Add the equipment's life cycle:** service and maintenance contracts, reagent purchases after equipment purchases, and asset registers. These answer the domain's real question, whether the equipment was still working a year later.
3. **Build a labelled taxonomy set and an evaluation set** of about 200 question-answer pairs with the domain experts.
4. **Run the Databricks pipeline** to switch on LLM lesson extraction, vector search and the monitoring job.
5. **Extend the scope configuration** to the next domain (e.g. medicines, or civil works). The pipeline already runs on an India roads scope and a three-country scope with no code changes.

10. Reproducing these numbers

```
python -m procurement_demo.run_all
python src/build_proposal.py
python src/build_deliverables.py
```

All figures come from the run outputs (`procurement_demo/output/health_lab/eval_results.json` and the Gold tables); none are typed by hand.